Spectral-Spatial Distribution Consistent Network Based on Meta-Learning for Cross-Domain Hyperspectral Image Classification

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***Abstract*— Cross-domain networks can solve the problem of insufficient labeled samples, especially for hyperspectral images (HSIs) where obtaining labeled samples is time-consuming and laborious. Most of the current methods rely on spatial information to achieve domain alignment, without considering the rich spectral information of HSIs. Furthermore, the methods based on a convolutional neural network (CNN) cannot get the spatial information of irregular image regions, resulting in poor classification results of object edges. Therefore, we design a spectral-spatial distribution consistent (SSDC) network based on meta-learning. First, to improve the feature extraction ability of the cross-domain classification model, we introduce a feature pre-extraction module, which uses the spectral attention mechanism and the alternating meta-learning method to obtain the general features of the source domain and the discriminative features of the target domain, so as to obtain the spectral weight matrix for subsequent processing. Second, we propose a spectral consistent module based on singular value decomposition (SVD), which increases the difference between different classes of features by penalizing the singular values of the feature matrix to achieve data distribution alignment in the spectral dimension. Finally, aiming at the low classification accuracy of irregular image regions, we propose a spatial consistent module to obtain non-local spatial topological information through stacked cross modules and graph sample and aggregate networks (graph SAGEs), which can reduce domain shift. The experiments of SSDC on four classical HSI datasets show that the proposed method can obtain competitive results with other methods based on CNN and cross-domain.**

***Index Terms*— Graph convolutional network (GCN), hyperspectral image (HSI) classification, meta-learning, singular value decomposition (SVD).**

I. Introduction

HYPERSPECTRAL image (HSI) is a remote sensing image containing rich spectral information, which can simultaneously image the object region by tens to hundreds of continuous and subdivided spectrum bands. HSIs have been used in many fields, such as precision agriculture [1], military reconnaissance [2], and environmental protection [3]. Among them, the accurate classification of land cover information has attracted much attention.

In the early studies of HSI classification (HSIC), most of them preferred to complete the classification process from a single dimension only, such as support vector machines (SVMs) [4] and principal component analysis (PCA) [5]. The above single-dimensional methods cannot effectively deal with the joint spatial and spectral information of HSI. Therefore, researchers proposed classification methods by combining spectral and spatial information. Wang and Liu [6] combined spatial and spectral information of each pixel with a composite kernel and used a weighted extreme learning machine (WELM) to achieve better classification. Li et al. [7] employed random forest to classify the independent spectral components and spatial features extracted by edge preserving filtering (EPF). However, the above traditional classification methods extract mostly hand-made shallow features with poor robustness, which cannot achieve high classification accuracy on unknown datasets.

Motivated by the successful application of deep learning (DL) in remote sensing, researchers discovered that the DL-based approaches to HSIC can extract shallow texture features and deep semantic features simultaneously, which produced better results. Chen et al. [8] first applied DL to HSIC and proposed a stacked auto-encoder (SAE) to obtain high-level features. However, this method stacks too many hidden layers, which is not conducive to training models by a large number of parameters. Therefore, Hu et al. [9] used a convolutional neural network (CNN) to HSIC and verified that CNN with lower parameters could greatly improve classification performance. Subsequently, CNN had become one of the most effective tools for extracting spatial and spectral features in HSI. Mei et al. [10] and Tang et al. [11] designed classification models based on 2D-CNN and 3D-CNN, respectively, but the detailed processing of edge pixels was poor. In order to further improve the classification of edge pixels in each region of HSI, Zhong et al. [12] and Bai et al. [13] proposed an end-to-end spectral-spatial residual network (SSRN) and a dual-channel dense network, respectively. However, CNN-based cannot effectively process

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the irregular image regions, resulting in poor classification results.

Therefore, the graph convolutional network (GCN) is proposed to make up for the deficiency of CNN, which can deal with irregular regions by constructing graph structure information based on images. Qin et al. [14] first applied GCN in HSIC, which used adjacent nodes of graph vertices to approximate convolution operation to obtain the classification result. This method can get the spatial features of irregular regions through graph topology structure and improve the classification performance. Hong et al. [15] proposed miniGCN, which adopted different fusion strategies for GCN and CNN to reduce the high computation complexity caused by numerous pixel nodes. However, directly taking each pixel as a graph node results in a long training time for the model. Zhang et al. [16] had found that the training time can be reduced by fusing a series of neighboring pixels with similar low-level features into a single superpixel. Sha et al. [17] proposed a graph attention network to obtain better classification results by assigning different weights to each pixel that compose the superpixels. Zhang et al. [18] proposed a spatial pooling GCN to preserve discriminative spatial features and reduce redundant pixels. Nevertheless, both CNN-based and GCN-based methods require training data and test data to be independent and identically distributed, which is hard to satisfy in real scenarios. Consequently, the cross-domain HSIC is proposed to break the strict assumption.

The cross-domain HSIC is to classify the unlabeled data in the target domain using the labeled data in the source domain and target domain. In general, the HSI dataset containing enough labeled data is called the source domain, and the dataset containing few labeled data is called the target domain. Typically, the data categories and distributions in the two domains are entirely different due to environmental changes, imaging equipment differences, and so on. Therefore, reducing the occurrence of domain shift is one of the key points in cross-domain problems. The general method is to map the samples from the source domain to the target domain by training the nonlinear mapping matrix to narrow the difference between two domains. Tzeng et al. [19] introduced a network based on adversarial training to make the marginal distribution of two domains consistent. Vu et al. [20] implemented the spatial alignment between different domains by minimizing the uncertainty of predicted results from the target data. Inspired by these methods, Li et al. [21] proposed a supervised heterogeneous transfer learning method for HSIC, which successfully implemented cross-domain HSIC for the first time. Wei et al. [22] constructed a low-rank classifier by aligning the subspace to obtain a robust domain-adaptation classification network. However, all the above methods require the target domain to provide sufficient labeled data to ensure the classification performance of the model, which does not effectively solve the problem of the lack of labeled data in real scenarios.

To solve the problem of insufficient label information, the method of few-shot learning (FSL) has attracted more attention in recent years. Meta-learning [23] is also used as a common tool for HSIC. Liu et al. [24] proposed a deep FSL (DFSL) method, which used the residual network to extract deep features. In the training phase, DFSL used meta-learning to learn general features, while in the testing phase, it used SVM and the nearest neighbor method to obtain the classification results of unlabeled data. Subsequently, Li et al. [25] combined FSL with cross-domain learning, and realized the category classification of each pixel in the target domain by alternating meta-learning between two domains.

However, the above methods mainly use the spatial features of HSIs to complete classification, and the CNN-based models are only applicable to regular regions. Therefore, to make full use of the rich spectral information of HSIs and improve the classification accuracy of irregular regions, we propose a spectral-spatial distribution consistent (SSDC) network to reduce the domain shift in the classification process. Fig. 1 shows the overall framework of SSDC uses spectral attention to obtain the spectral feature matrix with different weights and gets the singular values of the matrix by singular value decomposition (SVD). By penalizing the singular values to reduce the influence of complex channel information in deep semantic features on the difference among different categories, so as to realize the spectral distribution alignment. As for the irregular image regions, we use graph sample and aggregate network (graph SAGE) [26] and stacked cross modules to obtain nonlocal spatial topological features. To verify the validity of SSDC, we conduct experiments on four public HSI datasets. The experimental results show that SSDC outperforms other existing methods such as DCFSL [25] in the cross-domain HSIC.

The main contributions of SSDC are as follows.

1) In the proposed SSDC, domain alignment is performed from both spectral and spatial features, which are experimentally on four classical datasets for cross-domain HSIC.
2) The process of penalizing the singular values obtained by SVD reduces the influence of complex channel information in deep semantic features on category differences and helps the spectral distribution alignment.
3) The combination of graph SAGE and cross modules is used to obtain non-local spatial topology information of the image to reduce domain shift.

The remaining parts of this article are organized as follows. Section II reviews the related research in cross-domain and GCN. Section III introduced the proposed SSDC method. Experiment analysis of SSDC is discussed in Section IV. Finally, the conclusion is shown in Section V.

II. Preliminaries

A. Meta-Learning for Cross-Domain

Cross-domain classification is to classify unknown target domain information by using known source domain information. In cross-domain classification, in addition to how to reduce the distribution differences among multiple datasets, how to complete the classification with only a small amount of target domain information is also one of the difficulties. Meta-learning is a method that obtains a good hyperparameter by training other tasks first and then training a specific task. The main purpose is to enable the model to independently learn

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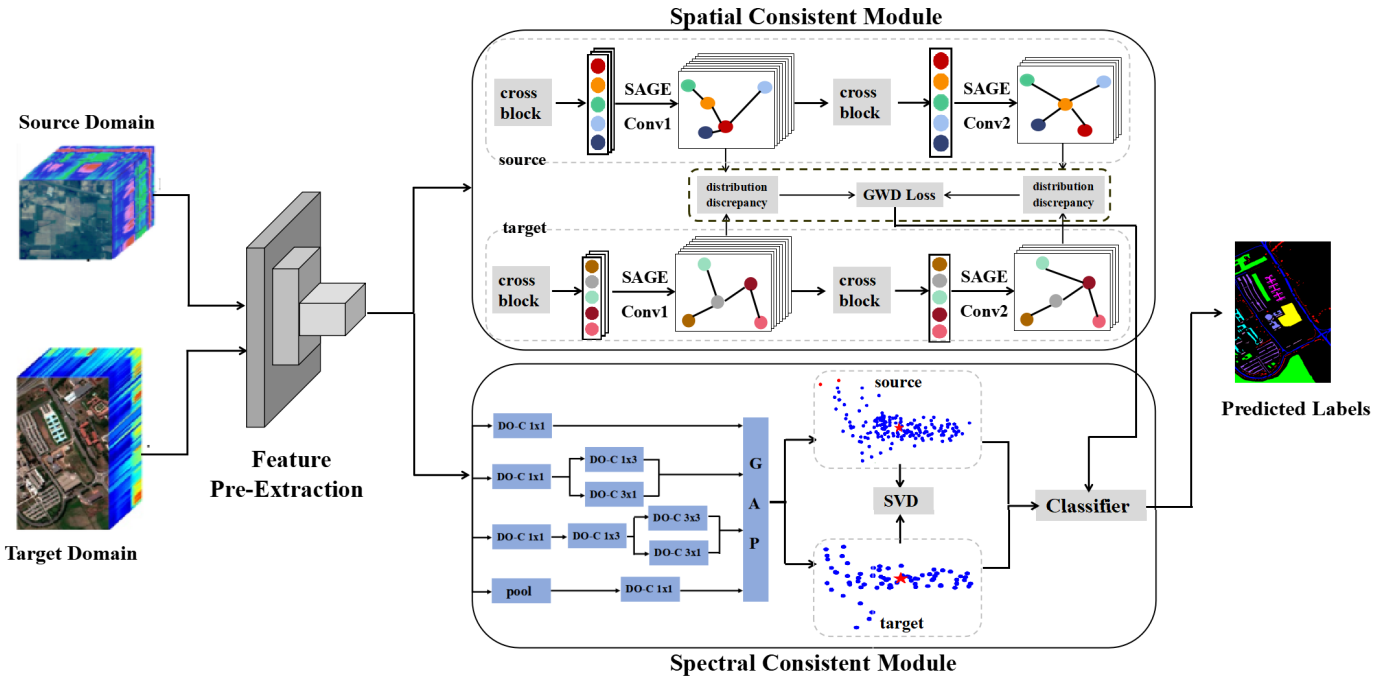

Fig. 1. Overall framework of SSDC includes three parts, namely feature pre-extraction module, spectral consistent module, and spatial consistent module. GAP denotes the global average pooling. DOCONV represents the depthwise over-parameterized convolution.

some new categories that have not been seen, that is, category information that has not appeared in the training process. As an autonomous learning method, meta-learning can complete the learning of new categories with only a few labeled samples, so it is chosen to add meta-learning to the cross-domain classification model to improve the classification performance. Cross-domain HSIC method based on meta-learning is to realize the category prediction of the target domain by alternating meta-learning in the source domain and target domain. Meta-learning is a C-way, K-shot problem. Specifically, it aims to randomly select C categories in the training set, each category contains K samples to form the support set, and the rest are used as the query set. Taking the source domain as an example, in order to simulate the meta-learning task, $C\times K$ labeled samples are randomly selected from the source domain to form the support set $S=(x_i, y_i)_{i=1}^{C\times K}$, and $C\times N$ unlabeled samples are selected to form the query set $Q=(x_j, y_j)_{j=1}^{C\times N}$ for training, where C is the number of classes in the target domain, x and y indicate data and corresponding label, respectively. In the target domain, meta-learning training is similar to the process in the source domain. Following equation is the loss function based on meta-learning

$$l_{\text{meta}}=E_{S,Q}\left[-\sum_{(x,y)\in Q}\log\left(\frac{\exp(-d(\phi(x),c_k))}{\sum_{k=1}^{K}\exp(-d(\phi(x),c_k))}\right)\right] \tag{1}$$

where $d(\cdot)$ represents the Euclidean distance function, $\phi(x)$ denotes the extracted feature in Q, and c_k indicates the feature of the k-th class in S. In this article, the meta-learning method is also used to achieve few-shot cross-domain HSI.

B. Graph Sampling Aggregation Network

CNN is based on the shared convolution kernel, and the feature matrix of the image is obtained by sliding windows. However, the classification performance of CNN for irregular image regions is poor. Therefore, GCN has been proposed to solve the above problem. The formula of GCN is generally expressed as

$$\mathbf{H}^{(l+1)}=\sigma\left(\widetilde{\mathbf{D}}^{-\frac{1}{2}}\widetilde{\mathbf{A}}\widetilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{H}^{(l)}\mathbf{W}^{(l)}\right) \tag{2}$$

where $\mathbf{H}^{(i)}$ represents the output of convolutional layer i, σ represents the activation function, $\widetilde{\mathbf{D}}$ is the degree matrix of $\widetilde{\mathbf{A}}$, expressed as $\widetilde{\mathbf{D}}_{ii}=\sum_j\widetilde{\mathbf{A}}_{ij}$. $\widetilde{\mathbf{A}}=\mathbf{A}+\mathbf{I}$, $\mathbf{A}$ is the adjacency matrix between nodes, and $\mathbf{I}$ is the identity matrix.

In order to select a more reasonable neighborhood size, the graph SAGE is proposed. Instead of training a separate embedding for each node, graph SAGE samples and aggregates the local neighborhood to obtain the feature representation of the node. Moreover, graph SAGE uses the feature information and structure information of nodes to map, which is more scalable than GCN and has a better performance for classification. At the same time, in the task of cross-domain HSIC, when new node information or another dataset is encountered, graph SAGE can dynamically aggregate the information of the new node, so as to better realize distribution alignment, reduce domain transfer and improve classification performance. The specific sampling strategy is to iterate random sampling on all first-order nodes connected to the central node. The aggregation method means that different aggregators are used to obtain features of the current nodes. Common aggregators include pooling aggregators and mean aggregators. The formula for

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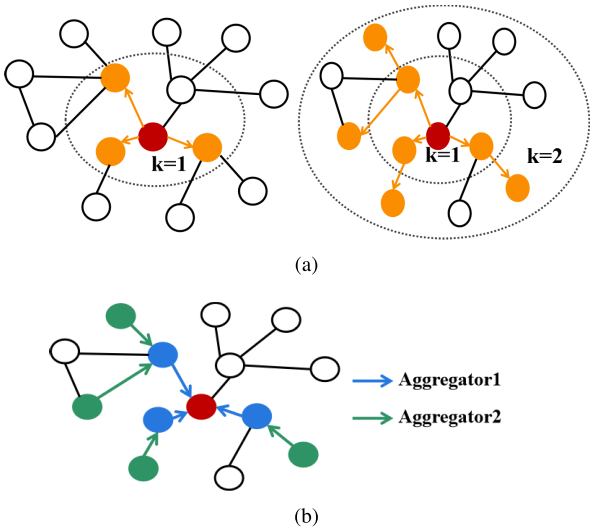

Fig. 2. Convolution operation of graph SAGE. (a) Sample neighborhood. (b) Aggregate and update feature information from neighbors.

the pooling aggregator is

$$h_v^k = \max\left(\sigma\left(\mathbf{W}_{\text{pool}} h_u^k + b\right)\right), \quad \forall u \in \mathbf{N}(v) \tag{3}$$

which may lose some information about neighboring nodes. The most widely used mean aggregator is expressed as

$$h_v^k = \sigma\left(\mathbf{W}_{\text{mean}} \cdot \text{mean}\left(h_v^{k-1} \cup h_u^{k-1}\right)\right), \quad \forall u \in \mathbf{N}(v). \tag{4}$$

Fig. 2 shows the process of graph SAGE, Fig. 2(a) represents the first- and second-order neighborhood of the red center node obtained by uniform sampling. Here, the number of pixels in the neighborhood set is determined artificially, and not all the first- or second-order pixels are regarded as neighborhood information. Fig. 2(b) denotes the information integration process of the red center node, that is, the feature representation of the central pixel is obtained by integrating the second- and first-order neighborhood information of the central pixel using the aggregation function. In this article, we select a mean aggregator for information integration.

III. Methodology

The overall framework of SSDC is shown in Fig. 1. SSDC is equipped with a feature pre-extraction module, a spectral consistent module (Spe), and a spatial consistent module (Spa). First, the training set of the classification network includes the source domain and the target domain, and then the two datasets are input into the feature pre-extraction module to extract features, which includes using $1 \times 1 \times 100$ convolution calculation to adjust the spectral dimension of the input features of the two datasets to the same, performing meta-learning alternately on the two domains to obtain the feature matrix with spectral attention weights and using a 3-D-residual network for feature preextraction as the input features of the subsequent two branches. Then, the obtained features are input into Spe and Spa, respectively. Spe uses the DOCONV-based multiscale module and SVD processing module to obtain the spectral consistency loss function. In Spa, the stacked cross modules and the graph SAGE are used to calculate the distribution

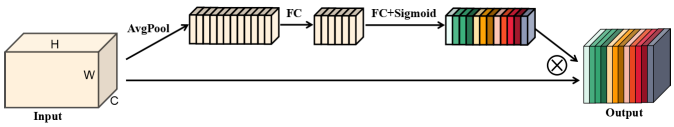

Fig. 3. Processing of spectral attention.

difference between the source domain and the target domain, and finally use all the loss functions to obtain the classification results.

A. Feature Pre-Extraction Module

The feature pre-extraction module is mainly composed of three parts, mapping module, spectral attention, and 3-D-residual network. The mapping module processes all input data to the same size by convolution and BatchNorm operations. On the premise of preserving the original data of the image as much as possible, considering the number of bands contained in the two datasets, we set the channel dimension of the output of the mapping module to 100.

Compared with other remote sensing images, HSI has a higher spectral dimension and greater redundancy, which affects the subsequent classification. In order to extract more important spectral information, consider selectively increasing the weight of the spectral containing more information. It has been proved that the squeeze-and-excitation (SE) networks [27] can fully consider the dependencies among channel dimensions, strengthen discriminative features and weaken other useless features. Inspired by this, we consider reducing spectral redundancy by calculating the spectral attention matrix. Fig. 3 shows the detailed process. The network preprocessed by spectral attention can increase the weight of features related to classification and weaken other unnecessary features. Specifically, GAP is used to map the spatial features to 1×1, and then two fully connected layers followed by a sigmoid nonlinear activation function are used to obtain the attention weights on the spectral dimension. Finally, the normalized weight matrix is multiplied with the input to obtain the weighted features. The specific mathematical expression is as follows:

$$f = \sigma(\mathbf{T}_2(\text{ReLU}(\mathbf{T}_1(\hat{\mathbf{X}})))) \tag{5}$$

where $\hat{\mathbf{X}}$ represents the feature after GAP, $\mathbf{T}_1$ and $\mathbf{T}_2$ denote two fully connected layers, and σ is the sigmoid activation function.

The 3-D-residual network consists of two residual blocks and a convolutional layer to map the weighted features into a unified metric space, while extracting deeper features as the input of the subsequent classification modules. We give specific parameter settings for 3-D-residual network in the experimental section. In order to learn the general features of the source domain and the discriminative features of the target domain, meta-learning is performed alternately on the two domains. After that, the obtained features are used as input to subsequent modules.

B. Spectral Consistent Module

Effective processing of spectral information and improving the difference among class features are the key to HSIC.

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To be able to reduce the spectral distribution shift among HSIs between the source and target domains, we design a spectral consistent module (Spe). The Spe contains a DOCONV-based multiscale module, SVD processing module, and classifier.

For image classification, using convolution kernels with different sizes can extract different ranges of context information, so as to improve the diversity of features and the classification performance of the model. Therefore, we design a DOCONV-based multiscale module with four branches, which can enhance the diversity of features by extracting features under different receptive fields. Since Spa is only involved in the training stage, in order to meet the requirements of spatial feature diversity in the test phase, the DOCONV-based multiscale module with spatial information added to the Spe is selected. The network performs meta-learning alternately on the two domains in the feature pre-extraction module, which results in a larger amount of data and a longer training time. Moreover, the graph structure is used to extract spatial features in the training phase, which leads to an increase in computational complexity. Due to the high computational complexity of extracting spatial features using graph structure, the training time of the model is longer. In order to speed up the convergence of the model, due to the influence of deep separable convolution, DOCONV is considered to replace traditional convolution operations, so as to ensure that the classification accuracy is not affected and the training speed of the model is accelerated. For the input feature of a specific channel, D_{mul} convolutional kernels are applied to window $M \times N$, usually D_{mul} is used as the depth multiplier. C_{in} and $D_{\text{mul}} \times C_{\text{in}}$ are the input and output feature channels, H and W are the height and width of the feature map. In DOCONV, $\mathbf{D}_{\text{mul}} \geq M \times N$ and $H \times W \gg C_{\text{out}}$. The convolutional result can be obtained according to the following formula:

$$\mathbf{O} = \mathbf{W} \otimes (\mathbf{D} \odot \mathbf{P}) \tag{6}$$

where $\odot$ and $\otimes$ denote depthwise convolution and traditional convolution, respectively. Here we define $\hat{\mathbf{p}} = \mathbf{D} \odot \mathbf{P}$. In the training phase, as for the networks containing DOCONV, (7) and (8) give the computational burden of the forward computation for CNN- and DOCONV-based optimization methods

$$\begin{aligned} \hat{\mathbf{P}} &\in \mathbb{R}^{D_{\text{mul}} \times (M \times N) \times C_{\text{in}} \times (H \times W)} \\ \mathbf{O} &\in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times H \times W \times D_{\text{mul}}} \end{aligned} \tag{7}$$

$$\begin{aligned} \hat{\mathbf{P}} &\in \mathbb{R}^{D_{\text{mul}} \times (M \times N) \times C_{\text{in}} \times C_{\text{out}}} \\ \mathbf{O} &\in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times H \times W \times (M \times N)}. \end{aligned} \tag{8}$$

It is obvious that the DOCONV-based method has less computation during training and the network converges faster. Fig. 4 shows the calculation process of DOCONV. The output is obtained by performing depthwise convolution and traditional convolution operations on the input. Depthiwse convolution operation means that the convolution is calculated channel by channel to ensure that the dimensions of the output and input are the same.

The deep semantic features processed by the multiscale module have stronger complexity, resulting in more complex channel information of the deep semantic features. Therefore, it is difficult to accurately distinguish the differences between

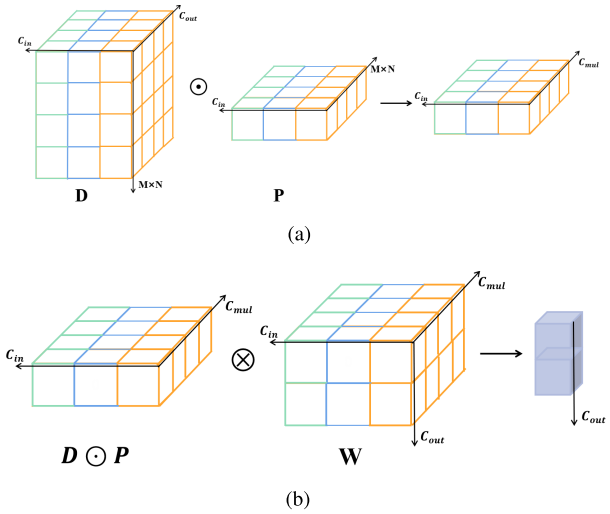

Fig. 4. Operations of DOCONV. (a) Depthwise convolution operation. (b) Traditional convolution operation.

various categories by directly using the fully connected layers. As a classical method of matrix decomposition, SVD decomposes the matrix into the left singular matrix, right singular matrix, and diagonal matrix composed of singular values, so that the eigenvalue information of the matrix is directly expressed by singular values, which greatly reduces the computational difficulty. In addition, HSI contains a high spectral dimension, which leads to the difficulty of extracting the eigenvalues of the feature matrix. Moreover, HSI contains a large number of bands with high correlation, resulting in high feature similarity of each category in the classifier. Therefore, in order to improve the diversity among the features of each class, SVD is used before the classification.

As an important matrix decomposition method in linear algebra, SVD is a generalization of eigen decomposition on any matrix. The mathematical formula for SVD of $\mathbf{A} \in \mathbb{R}_r^{m \times n}$ is as follows:

$$\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^T = \mathbf{U} \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & 0 \\ & & \ddots & \\ & & & \sigma_r \\ & 0 & & 0 \end{bmatrix} \mathbf{V}^T \tag{9}$$

where $\Sigma \in \mathbb{R}^{m \times n}$ and the elements $\sigma_i (i = 1, 2, \ldots, r)$ on the diagonal of Σ are singular values. $\mathbf{U} \in \mathbb{R}^{m \times m}$ is the left singular matrix and $\mathbf{A}\mathbf{A}^T = \mathbf{U}\Sigma^2\mathbf{U}^T$. The columns of $\mathbf{U}$ form a set of basis vectors for orthogonal input of $\mathbf{A}$ and these vectors are the eigenvectors of $\mathbf{A}\mathbf{A}^T$. $\mathbf{V} \in \mathbb{R}^{n \times n}$ is right singular matrix and $\mathbf{A}^T\mathbf{A} = \mathbf{V}\Sigma^2\mathbf{V}^T$. The columns of $\mathbf{V}$ form a set of basis vectors for orthogonal input of $\mathbf{A}$ and these vectors are the eigenvectors of $\mathbf{A}^T\mathbf{A}$.

In the decomposition process, the larger singular values obtained indicate that the feature vector contains more channel information. Therefore, the influence of complex channel information on the difference in category features is reduced by minimizing the singular values. To increase the penalty strength, we use the sum of squared singular values in the loss function. Here, the spectral consistency loss function l_{svds}

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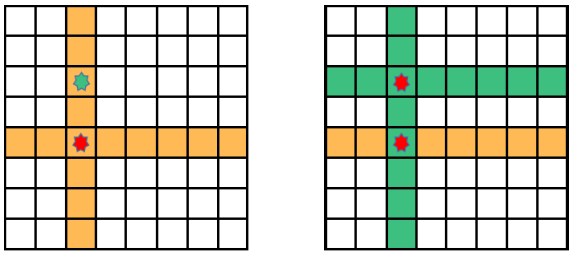

(a) (b)

Fig. 5. Non-local spatial information based on stacked cross modules. (a) Local information for the same row and column. (b) Non-local information.

in the source domain is given

$$l_{\text{svds}} = \sum_{i=1}^{r} \sigma_i^2. \tag{10}$$

Similarly, the spectral consistency loss function in the target domain l_{svdt} is set the same as that in the source domain. After experimental analysis, the weights of l_{svds} and l_{svdt} are set to 0.45 and 0.55, respectively. Finally, the classification results are obtained by using the classifier.

C. Spatial Consistent Module

At present, most CNN-based networks use the local spatial information of the image for classification, and the classification effect is poor for irregular image regions. Since the shape of the convolution kernel is regular or rectangular, and the two matrices of the convolution calculation have the same size, the CNN-based network can only extract features in the corresponding regular region, which makes it difficult to extract spatial information of irregular data structures [14], [28]. Specifically, the local spatial features cannot extract the relationships among the long-distance pixels of the same category, which leads to the missing or misclassification of the pixels. With the continuous expansion of the receptive field, the CNN-based network is easily affected by other irrelevant information during the training phase, and the graph topology can not only solve the above problems, but also effectively deal with irregular data structures.

In addition, in cross-domain classification learning, the model needs to learn as much information as possible from the source domain and the target domain to better simulate the data distribution, so as to achieve alignment and reduce the occurrence of domain transfer. Therefore, in order to efficiently extract some irregular image regions contained in HSIs and improve classification accuracy, we construct a spatial consistent module (Spa) in cross-domain learning. Spa uses stacked cross modules to extract the global spatial features, and graph SAGE acquires the spatial topology of the image.

In order to obtain the global and local spatial information of the image at the same time, we build a cross-module inspired by the criss-cross network (CCNet). First, a single cross module is used to obtain local information, and then another cross module is used to extract the global spatial information of the image. CCNet has proved that the stacked cross module can get non-local features. Fig. 5 shows the structure of the cross-module. The cross-module is composed of all pixels in the same row and column to form the neighborhood of the center pixel. For the input feature map $\mathbf{X} \in \mathbb{R}^{C\times W\times H}$, where C, W, and H represent channels, width, and height of features, respectively, $\mathbf{Q} \in \mathbb{R}^{C'\times W\times H}$, $\mathbf{K} \in \mathbb{R}^{C'\times W\times H}$, and $\mathbf{V} \in \mathbb{R}^{C\times W\times H}$ are obtained through three convolution layers, where C' represents channel numbers after dimensionality reduction. Then, the affinity is made to $\mathbf{Q}$ and $\mathbf{K}$. The specific calculation process is as follows:

$$D_{i,u} = \mathbf{Q}_u \Omega_{i,u}^T, \quad i \in [0, 1, \ldots, H + W - 1], \quad u \in [0, 1, \ldots, H \times W] \tag{11}$$

where $\mathbf{Q}_u \in \mathbb{R}^{C'}$ represents each position u of $\mathbf{Q}$ on the spatial dimension and $\Omega_u \in \mathbb{R}^{(H+W-1)\times C'}$ represents the neighborhood set of the same row and column of position u on $\mathbf{K}$. $\mathbf{D} \in \mathbb{R}^{(H+W-1)\times(H\times W)}$ is the correlation matrix.

The matrix $\mathbf{A} \in \mathbb{R}^{(H+W-1)\times(H\times W)}$ is obtained by using softmax on the channel dimension of $\mathbf{D}$, where the channel dimension is the $(H + W - 1)$ dimension, indicating the correlation between a pixel and other pixels in the vertical and horizontal direction. In the same way, the vector $\mathbf{V}_u \in \mathbb{R}^C$ and the set $\Phi_u \in \mathbb{R}^{(H+W-1)\times C}$ can be obtained from $\mathbf{V}$. Finally, the aggregation operation is used to obtain the new feature map

$$\mathbf{X}'_u = \sum_{i=0}^{H+W-1} \mathbf{A}_{i,u} \Phi_{i,u} + \mathbf{X}_u \tag{12}$$

where $\mathbf{X}_u$ and $\mathbf{X}'_u$ represent the feature vectors of the original and new feature map at position u, respectively. Since the global spatial information leads to the expansion of the receptive field, the network is susceptible to the influence of irrelevant information in the training phase. At the same time, to improve the classification accuracy of irregular image regions, the feature matrix information is extracted from the topological structure of the graph. Specifically, the graph SAGE based on the mean aggregator to aggregate neighborhood node information is chosen. Graph SAGE can obtain the first-order neighborhood of each pixel through repeated uniform sampling, which improves the correlation between each pixel in the neighborhood, and then improves the classification performance. By applying stacked cross modules and graph SAGE, the non-local spatial topology of the feature map is obtained.

We use the Gromov–Wasserstein distance (GWD) [29] to measure the spatial distribution difference between the two domains. $P(\mathbf{X}_s)$ and $P(\mathbf{X}_t)$ represent the data distributions of the source domain and target domain. The spatial consistency loss function is expressed as

$$l_{\text{gwd}}[P(\mathbf{X}_s), P(\mathbf{X}_t)] = \inf_{\gamma \in \Pi[P(\mathbf{X}_s), P(\mathbf{X}_t)]} \mathop{\mathbb{E}}_{\substack{(\mathbf{X}^s, \mathbf{X}^t)\sim\gamma \\ (\hat{\mathbf{X}}^s, \hat{\mathbf{X}}^t)\sim\gamma}} \left[L\left(\mathbf{X}_i^s, \mathbf{X}_j^t, \hat{\mathbf{X}}_i^s, \hat{\mathbf{X}}_j^t\right) \right] \tag{13}$$

$$L\left(\mathbf{X}_i^s, \mathbf{X}_j^t, \hat{\mathbf{X}}_i^s, \hat{\mathbf{X}}_j^t\right) = \left\| c\left(\mathbf{X}_i^s, \hat{\mathbf{X}}_i^s\right) - c\left(\mathbf{X}_j^t, \hat{\mathbf{X}}_j^t\right) \right\| \tag{14}$$

where $c(\mathbf{X}, \hat{\mathbf{X}})$ indicates the cosine similarity distance between nodes. $L(\cdot)$ represents the in-graph structural similarity between two pairs of nodes $(\mathbf{X}_i^s, \hat{\mathbf{X}}_i^s)$ and $(\mathbf{X}_i^t, \hat{\mathbf{X}}_i^t)$. It should

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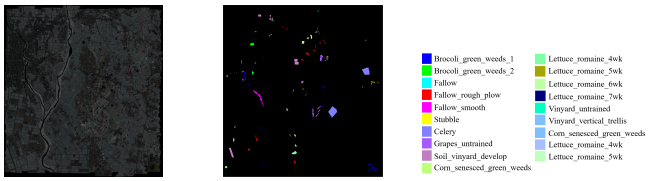


(a) (b)

Fig. 6. Chikusei dataset. (a) False-color image. (b) Ground-truth map.

TABLE I
EACH CLASS AND CORRESPONDING SAMPLE NUMBER OF CHIKUSEI DATASET IN SOURCE DOMAIN

Class	Class Name	Data
1	Water	2845
2	Bare soil (school)	2859
3	Bare soil (park)	286
4	Bare soil (farmland)	48525
5	Natural plants	4297
6	Weeds in farmland	1108
7	Forest	20516
8	Grass	6515
9	Rice field (grown)	13369
10	Rice field (first stage)	1268
11	Row crops	5961
12	Plastic house	2193
13	Manmade (non-dark)	1220
14	Manmade (dark)	7664
15	Manmade (blue)	431
16	Manmade (red)	222
17	Manmade grass	1040
18	Asphalt	801
19	Paved ground	145
	TOTAL	77,592

be noted that Spa only participates in the training process, not in the testing process.

IV. EXPERIMENTS

A. Data Description

In this section, we use five common HSI datasets for experiments, namely, the Chikusei [30], the Indian Pines (IP), the Salinas (SA), the University of Pavia (PU), and the Pavia Center (PC). The Chikusei has the largest number of land cover categories and is used as the source domain dataset, while the others are used as the target domain datasets. The details of each dataset are given here.

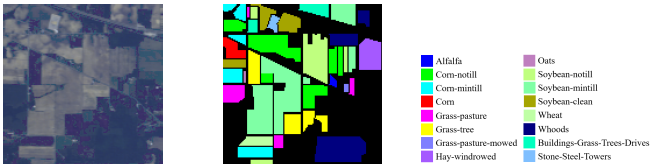
1) Source Domain: The Chikusei dataset was gathered by Hyperspectral Visible/Near-Infrared Cameras (Hyperspec-VNIR-C) in Chikusei, Ibaraki, Japan. It contains 2571 × 2335 pixels. There are 19 classes representing different types of land cover. And its spatial resolution is 2.5 m/pixel. It has 128 spectral bands, which range from 343 to 1018 nm. The false-color image and corresponding ground-truth map can be seen in Fig. 6. Table I shows the samples of the Chikusei dataset.

2) Target Domain: In order to verify the validity and generalization ability of SSDC, we select four datasets to construct target domain data, namely, IP, SA, PU, and PC.

The IP dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in Northwest Indian. It contains 16 classes and has 145 × 145 pixels. Its spatial resolution is 20 m/pixel. It consists of 200 spectral bands, which range from 400 to 2500 nm. The false-color image and corresponding ground-truth map can be seen in Fig. 7. Table II shows the samples of the IP dataset.

TABLE II
EACH CLASS AND CORRESPONDING SAMPLE NUMBER OF INDIAN PINES DATASET IN TARGET DOMAIN

Class	Class Name	Data
1	Alfalfa	46
2	Corn-notill	1428
3	Corn-mintill	830
4	Corn	237
5	Grass-pasture	483
6	Grass-trees	730
7	Grass-pasture-mowed	28
8	Hay-windrowed	478
9	Oats	20
10	Soybean-notill	972
11	Soybean-mintill	2455
12	Soybean-clean	593
13	Wheat	205
14	Woods	1265
15	Buildings-Grass-Trees-Drives	386
16	Stone-Steel-Towers	93
	TOTAL	10,249



(a) (b)

Fig. 7. IP dataset. (a) False-color image. (b) Ground-truth map.

TABLE III
EACH CLASS AND CORRESPONDING SAMPLE NUMBER OF SALINAS DATASET IN TARGET DOMAIN

Class	Class Name	Data
1	Brocoli_green_weeds_1	2009
2	Brocoli_green_weeds_2	3726
3	Fallow	1976
4	Fallow_rough_plow	1394
5	Fallow_smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes_untrained	11271
9	Soil_vinyard_develop	6203
10	Corn_senesced_green_weeds	3278
11	Lettuce_romaine_4wk	1068
12	Lettuce_romaine_5wk	1927
13	Lettuce_romaine_6wk	916
14	Lettuce_romaine_7wk	1070
15	Vinyard_untrained	7268
16	Vinyard_vertical_trellis	1807
	TOTAL	54,129

The SA dataset was captured by an AVIRIS sensor over Salinas Valley, CA, USA. It consists of 512 × 217 pixels. Its spatial resolution is 3.7 m/pixel. It has 204 spectral bands

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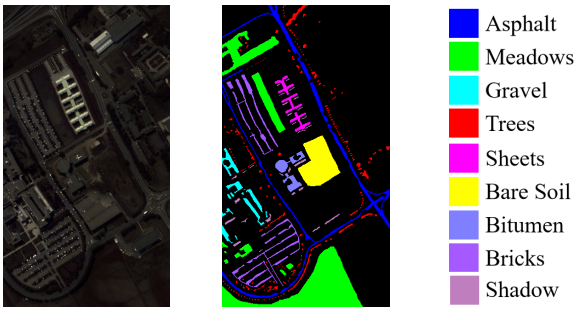

Fig. 8. PU dataset. (a) False-color image. (b) Ground-truth map.

TABLE IV
EACH CLASS AND CORRESPONDING SAMPLE NUMBER OF UNIVERSITY OF PAVIA DATASET IN TARGET DOMAIN

Class	Class Name	Data
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Sheets	1345
6	Bare soil	5029
7	Bitumen	1330
8	Bricks	3682
9	Shadow	947
	TOTAL	42,776

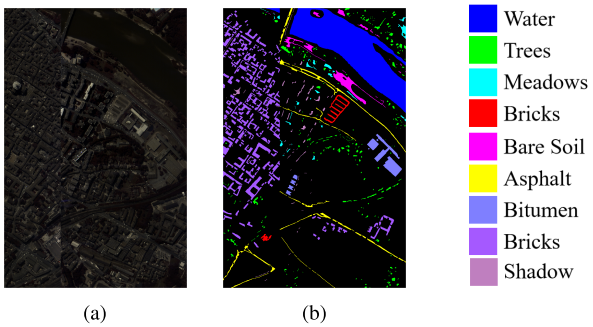

Fig. 9. PC dataset. (a) False-color image. (b) Ground-truth map.

and ranges from 400 to 2500 nm. The ground truth contains 16 different classes. Table III shows the samples of SA dataset.

The PU dataset contains 610×340 pixels. Its spatial resolution is 1.3 m/pixel. It has 103 bands that range from 430 to 860 nm. It was captured by the Reflective Optics Spectrographic Image System (ROSIS). It contains nine classes. The false-color image and corresponding ground-truth map can be seen in Fig. 8. Table IV shows the samples of the PU dataset.

The PC dataset was also gathered by ROSIS sensor over Pavia, Northern Italy. It consists of 102 spectral bands and ranges from 430 to 860 nm. It has 1906×715 pixels. Its spatial resolution is 1.3 m/pixel. It includes nine classes. The false-color image and corresponding ground-truth map can be seen in Fig. 9.

TABLE V
DETAILED PARAMETERS OF EACH MODULE

Module	Layer		Output Size
Input			$B \times M \times 9 \times 9$
Feature Pre-Extraction	Mapping		$B \times 100 \times 9 \times 9$
	Channel Attention		$B \times 100 \times 9 \times 9$
	3D Residual Block	Conv	$B \times 8 \times 100 \times 9 \times 9$
		Conv	$B \times 8 \times 100 \times 9 \times 9$
		Conv	$B \times 8 \times 100 \times 9 \times 9$
		shortcut	$B \times 8 \times 100 \times 9 \times 9$
	Max Pooling		$B \times 8 \times 25 \times 9 \times 9$
	3D Residual Block	Conv	$B \times 16 \times 25 \times 9 \times 9$
		Conv	$B \times 16 \times 25 \times 9 \times 9$
		Conv	$B \times 16 \times 25 \times 9 \times 9$
		shortcut	$B \times 16 \times 25 \times 9 \times 9$
	Max Pooling		$B \times 16 \times 7 \times 5 \times 5$
	Conv		$B \times 32 \times 5 \times 5 \times 5$
Spectral Consistency	DOCONV		$B \times 128 \times 5 \times 5$
	DOCONV		$B \times 64 \times 5 \times 5$
	multi-scale		$B \times 128 \times 1 \times 1$
	Classifier		$B \times C$
Spatial Consistency	Cross		$B \times 160 \times 5 \times 5$
	SAGE		$B \times 3200$
	Cross		$B \times 128 \times 5 \times 5$
	SAGE		$B \times 1600$
	multi-sacle		$B \times 128 \times 1 \times 1$

1. M represents the number of HSI bands.
2. C represents the number of classes.
3. B represents batch-size.

B. Experimental Settings

All experiments are conducted on an Nvidia GeForce RTX 3090 graphics processing unit (GPU) with 24 GB memory. The detailed parameter setting in the training and testing phase is given. The number of land cover categories in the source domain is 19, which is larger than that in the target domain. In the training stage, according to the number of categories in the target domain, the same number of categories are randomly selected from the Chikusei dataset to form the training data in the source domain. The specific parameters of the 3-D-residual network are shown in Table V, where Conv is the traditional convolution operation. In the DOCONV-based multiscale module, the first, second, third, and fourth represent all operations of corresponding branches, respectively.

In addition to the loss functions mentioned above, pseudo-labeled data are obtained by generative adversarial network (GAN) before training to alleviate the problem of insufficient labeled data in the target domain. To reduce the error accumulation caused by pseudo-labeled data, we design a domain discriminator $\mathbf{D}$. Let $f = \mathbf{f}(x)$ denote the features extracted by SSDC, and $g = \mathbf{g}(x)$ represent the classification result. The domain adversarial loss l_{adv} is

$$l_{\text{adv}} = \min_{\mathbf{D}} \max_{\mathbf{F},\mathbf{G}} = - \mathop{\mathbb{E}}_{x_i^s \sim P(\mathbf{X}_s)} \log\left[\mathbf{D}\left(f_i^s, g_i^s\right)\right] - \mathop{\mathbb{E}}_{x_j^t \sim P(\mathbf{X}_t)} \log\left[1 - \mathbf{D}\left(f_j^t, g_j^t\right)\right] \tag{15}$$

where $\mathbf{D}(\cdot)$ is the probability that the predicted sample x belongs to the source domain and $1 - \mathbf{D}(\cdot)$ is the probability

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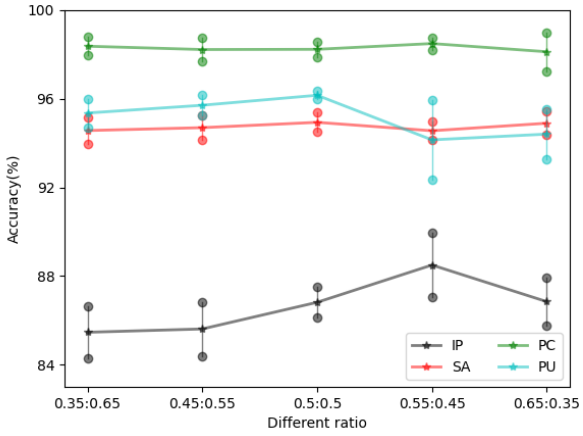


Fig. 10. Comparison the classification accuracy of different proportions of l_{svds} and l_{svdt} on each dataset.

TABLE VI
NUMBER OF LABELED AND UNLABELED SAMPLES FOR EACH CATEGORY ON EACH DATASET

	IP	SA	PU	PC
support set	9	9	12	6
query set	18	18	24	12

that the predicted sample belongs to the target domain. In the training stage, the meta-learning method is adopted to deal with the few-shot cross-domain HSIC, and the learning is conducted alternately in the two domains. The loss function is shown in (1). The final complete loss function of SSDC is

$$L = l_{\text{meta}} + l_{\text{adv}} + l_{\text{gwd}} + \alpha * l_{\text{svds}} + \beta * l_{\text{svdt}}. \quad (16)$$

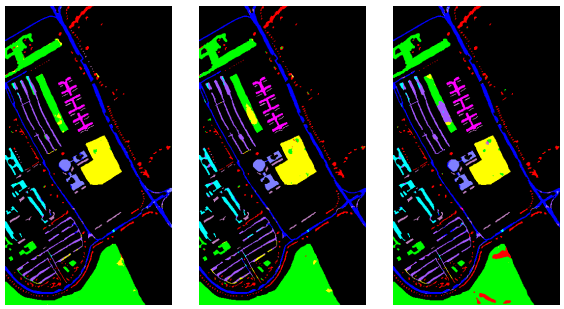
For parameters α and β in the loss function, the classification accuracy of different proportions is shown in Fig. 10. We finally choose $\alpha = 0.55$ and $\beta = 0.45$.

The SSDC is trained for Spe and **D** by Adam optimizer with 0.001 learning rate, and for Spa by SGD optimizer with 0.01 learning rate. All convolution kernels are initialized with Xavier normalization. The number of training iterations is set to 5000 for ten iterations, which is enough to train the network. In meta-learning, the proportion of labeled and unlabeled samples in all datasets is 1 : 2. The specific numbers in each dataset are shown in Table VI, which represent the samples in each category. The labeled sample in the target domain is pseudo-label samples generated by GAN. In the testing phase, we use the trained model to obtain the features, and then obtain the classification results by clustering using the k-nearest neighbor (KNN).

To further verify the classification performance of the proposed SSDC, we compare SSDC with supervised methods SVM [4], 2-D CNN [31], and SSRN [12], semisupervised method SSCNN [32], GCN-based methods GCN [33] and FuNet_C [15], few-shot methods DFSL+NN [24] and DFSL+SVM [24], cross-domain methods DCFSL [25] and TSTnet [34]. In order to fairly compare the performance of each algorithm, the amount of training set for all methods except supervised methods is set to be 1.5% of the total amount of data. For supervised methods, we set it to 3%.

TABLE VII
CLASSIFICATION ACCURACY OF ALL CATEGORIES (%), OA, AA, AND KAPPA IN DIFFERENT SOURCE DOMAINS ON THE PU AND PC DATASETS. THE BEST VALUES ARE HIGHLIGHTED IN BOLD FONT

Classes	PU_{IP}	PU_{SA}	PU_C	PC_{IP}	PC_{SA}	PC_C
1	92.43	93.10	**94.79**	99.87	99.39	**100.00**
2	89.46	92.49	**97.77**	93.24	91.79	**93.66**
3	90.22	**94.80**	92.72	95.55	**98.64**	92.50
4	96.94	96.55	**97.04**	99.03	94.13	**99.03**
5	100.00	**100.00**	99.62	84.34	**89.64**	87.14
6	98.66	94.91	**99.02**	**97.93**	89.94	95.40
7	90.89	93.19	**93.49**	87.55	**95.48**	87.67
8	81.47	81.68	**89.67**	97.62	95.92	**98.93**
9	98.92	99.57	**100.00**	**98.28**	95.76	97.44
OA(%)	91.46	92.75	**96.44**	97.33	96.60	**97.68**
AA(%)	93.22	94.03	**96.01**	**94.82**	94.52	94.64
Kappa(×100)	88.9080	90.5042	**95.2866**	96.2262	95.2131	**96.7176**



(a) (b) (c)

Fig. 11. Classification results obtained by different source domains for the PU and PC datasets. (a) PU_{IP}. (b) PU_{SA}. (c) PU_C.

The training data size in all algorithms is 9×9. We select the overall accuracy (OA), average accuracy (AA), kappa coefficient (Kappa), and accuracy of each class for evaluation. To clarify, we define the class that contains more samples as a large class, and the class that contains fewer data as a small class.

C. Different Source Domain

In order to further verify the performance of SSDC, different datasets are selected as source domains for experiments. Table VII shows the classification results of PU and PC datasets by SSDC in different source domains, where PU_{IP}, PU_{SA}, and PU_C represent the results of the PU dataset using IP, SA, and Chikusei as source domains, respectively, and the representation of PC_{IP}, PC_{SA}, and PC_C is similar.

In the PC dataset, the classification accuracy of PC_{IP} and PC_C is roughly the same, while PC_{SA} is the opposite of them. In the small categories of Meadows and Bitumen, the accuracy of PC_{SA} is 3%, 6.2%, and 7.9%, 7.8% higher than that of PC_{IP} and PC_C, respectively. In the large categories of Bricks and Asphalt, the accuracy of PC_{IP} and PC_C is 4.9%, 4.9%, and 8%, 5.5% higher than that of PC_{SA}. The results show that in cross-domain learning, even if the datasets in the source domain are from the same sensor, such as IP and SA datasets, the classification results of the same target domain are different.

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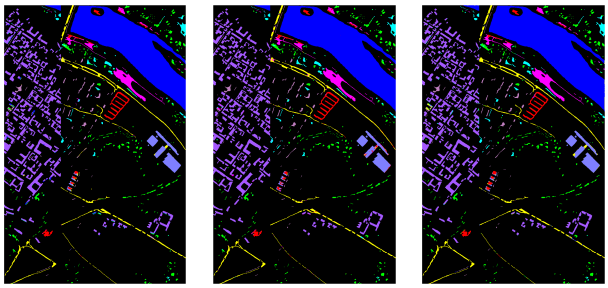
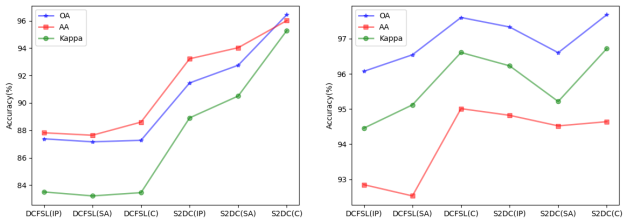


(a) (b) (c)

Fig. 12. Classification results obtained by different source domains in the PC datasets. (a) PC_{IP}. (b) PC_{SA}. (c) PC_C.



(a) (b)

Fig. 13. Classification results of different source domain datasets. (a) Results on PU dataset. (b) Results on PC dataset.

In the PU dataset, the OA of PU_C is 4.9% and 3.7% higher than that of PU_{IP} and PU_{SA}, respectively, and the accuracy of each category is almost the best. In the PC dataset, the OA of PC_C is 0.35% and 1% higher than that of PC_{IP} and PC_{SA}, respectively. The specific visualization results are shown in Figs. 11 and 12. It can be seen that the classification results of PU_C and PC_C are better. The reason is that the Chikusei dataset contains more categories than the IP and SA datasets, which enables the model to learn more diverse information during the cross-domain training phase. Therefore, SSDC chooses the Chikusei dataset as the source domain to classify other datasets.

The generalization ability of the model is one of the key factors to evaluate its performance of cross-domain methods. As shown in Fig. 13, OA, AA, and Kappa are given, where SA, C, and IP indicate that the SA, Chikusei, and IP datasets are used as the source domain to classify the different target domains, respectively. Compared with DCFSL, SSDC can obtain better classification results no matter which dataset is selected as the source domain, and OA varies little among different datasets. As shown in Fig. 13(a), in the PC dataset, the OA gap of DCFSL-based methods is about 2%, while that of the SSDC-based is less than 1%, which also verifies that SSDC has better robustness and generalization. The experimental results show that SSDC can obtain better classification results and stronger robustness by extracting spectral and spatial features simultaneously.

D. Impact of Data Distribution Consistency

Since Spa is only used for model training, in order to better verify the effectiveness of SSDC, Spe, and Spa are separately analyzed experimentally. Ablation experiments are performed on IP, SA, PU, and PC datasets of different sensors to investigate the effect of Spe and Spa in reducing cross-domain HSIC domain transfer. $SSDC_{Spe}$ represents SSDC with only spectral consistent module, and $SSDC_{Spa}$ represents SSDC with only spatial consistent module.

TABLE VIII
EFFECT OF SPA AND SPE MODULES ON OVERALL ACCURACY [OA(%)] IN EACH DATASET

	DCFSL	$SSDC_{Spe}$	$SSDC_{Spa}$	$SSDC$
IP	77.00	84.08	84.69	**86.77**
SA	93.42	93.98	94.59	**94.97**
PU	90.28	94.24	95.10	**95.76**
PC	97.26	98.04	98.23	**98.29**

TABLE IX
TRAINING AND TEST TIME OF SSDC WITH DIFFERENT CONVOLUTION ON EACH DATASET

		$SSDC_{DOCONV}$	$SSDC_{Conv}$
IP	train time(s)	10878.37697	21890.38672
	test time(s)	2.28080	3.06606
	OA(%)	84.53	86.01
SA	train time(s)	20074.76306	26604.30782
	test time(s)	11.59880	12.13693
	OA(%)	94.59	94.56
PU	train time(s)	18266.23067	22773.88462
	test time(s)	12.36746	13.90975
	OA(%)	95.55	95.52
PC	train time(s)	17884.77895	24850.24828
	test time(s)	67.58060	72.60146
	OA(%)	97.89	97.90

As shown in Table VIII, using either Spe or Spa alone can improve classification performance compared to DCFSL. This indicates that both spectral distribution alignment and non-local spatial topology information are important for classification accuracy. We can also see that SSDC gets better results compared to $SSDC_{Spe}$ and $SSDC_{Spa}$. In the IP dataset, the OA of SSDC is 2.7% and 2.1% higher than that of $SSDC_{Spe}$ and $SSDC_{Spa}$, respectively. The OA of SSDC is also higher on the other datasets. The results are also consistent with the theoretical analysis. Naturally, SSDC selects both Spe and Spa for classification.

E. Different Convolution

To obtain spatial information of irregular image regions, SSDC uses graph SAGE in Spa. Inevitably, graph SAGE increases the training time significantly. In order to speed up the convergence of the model, we use DOCONV instead of the traditional convolution operation. It can be seen from Table IX that the training time of $SSDC_{DOCONV}$ is reduced by 50%, 24.5%, 19.7%, and 28%, respectively, compared with $SSDC_{Conv}$ on the IP, SA, PU, and PC datasets. In the SA and PU datasets, the OA of the model is improved by 0.03%. In the PC dataset, the OA decreases by 0.01%, which is almost negligible. The experimental results show that DOCONV can

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TABLE X
CLASSIFICATION ACCURACY OF ALL CATEGORIES (%), OA, AA, AND KAPPA BY DIFFERENT METHODS ON THE INDIAN PINES DATASET

Classes	SVM	2-D CNN	SSRN	SSCNN	GCN	FuNet_C	DFSL+NN	DFSL+SVM	DCFSL	TSTnet	SSDC
1	97.30	100.00	92.86	97.62	98.81	100.00	98.81	100.00	**100.00**	99.17	99.64
2	20.16	68.49	80.43	26.50	55.39	68.42	44.37	42.39	63.95	39.31	**81.17**
3	31.91	73.81	5.54	74.47	61.49	61.45	47.33	42.32	65.76	40.45	**85.38**
4	20.18	92.85	**100.00**	78.39	95.89	96.19	77.63	76.41	89.95	79.51	98.08
5	67.51	90.61	86.09	58.06	83.37	**91.76**	79.21	76.99	85.66	76.45	91.72
6	89.04	98.78	**99.49**	60.21	89.14	95.13	90.68	87.41	92.74	93.65	95.87
7	100.00	100.00	100.00	63.33	89.28	100.00	100.00	100.00	100.00	88.12	**100.00**
8	94.03	96.74	**99.86**	85.36	89.28	97.03	86.88	91.59	95.72	98.48	98.78
9	100.00	100.00	100.00	66.67	100.00	100.00	100.00	100.00	100.00	96.40	**100.00**
10	54.72	81.34	74.81	72.12	77.92	75.37	70.65	64.08	76.38	100.00	**83.38**
11	58.63	69.10	71.88	17.80	70.05	70.21	61.06	62.18	66.60	67.74	**79.13**
12	24.83	78.96	59.07	42.84	46.49	69.91	52.81	38.26	66.90	53.64	**79.77**
13	98.98	100.00	100.00	99.82	99.82	100.00	99.29	100.00	**100.00**	49.46	99.36
14	78.98	93.13	**96.69**	88.18	68.81	72.31	83.51	89.82	94.79	97.49	96.01
15	34.22	92.21	64.13	95.20	72.74	87.32	70.56	69.47	86.59	84.96	**94.54**
16	95.24	99.56	99.56	33.33	100.00	100.00	98.67	**100.00**	99.56	100.00	98.80
OA(%)	55.06	80.86	75.42	53.27	70.96	76.11	66.86	65.60	77.00	73.99	**86.77**
AA(%)	66.61	89.72	83.15	66.24	81.82	86.57	78.84	77.56	86.54	84.78	**92.60**
Kappa($\times$100)	49.1967	78.3409	72.2347	48.6859	67.1447	72.9350	62.6902	61.2967	73.9743	71.5281	**84.9763**

not only ensure the classification results but also shorten the calculation time. Therefore, SSDC chooses DOCONV instead of traditional convolution.

F. Compared With Supervised and Semisupervised Methods

Based on the radial basis kernel function, SVM maps nonlinear data to a linearly separable space. Two-dimensional CNN extracts feature through the convolution kernel of size 3×3. SSRN uses the residual network to extract spectral and spatial features. SSCNN designs supervised and unsupervised loss functions for classification. As shown in Table X, in the IP dataset, OA of 2-D CNN and SSRN increases by 25% and 20% than SVM, respectively. This verifies that compared with traditional methods, DL-based methods can obtain better results. It can also be seen that the performance of supervised methods is better than that of the semisupervised method SSCNN, which indicates that the fewer labeled samples, the worse learning ability of the model. Tables XI–XIII have the similar results.

As shown in Tables XI–XIII, whether supervised or semisupervised methods, SSDC based on graph SAGE has better results. In the IP dataset, the OA of SSDC is 31%, 6%, 11%, and 33.5% higher than that of SVM, 2-D CNN, SSRN, and SSCNN, respectively. For the large category of Soybean-mintill, SSDC obtains the highest accuracy of 79.13%, which is 20.5%, 10%, and 7.3% higher than that of supervised methods SVM, 2-D CNN, and SSRN, respectively, and improves the classification performance on large classes. In the small categories of Oats and Grass-pasture-mowed, SSDC also gets the highest accuracy of 100.00%. In other datasets, even if SSDC does not get the highest accuracy in large or small classes, the difference is not significant. In the PC dataset, SSDC is 0.14% lower on the large category Water and 0.1% lower on the small category Bricks compared to the best result. The above analysis shows that using graph SAGE to extract spatial topological features can improve the classification accuracy of the model for large classes without affecting the classification results of small classes.

G. Compared With GCN-Based Methods

GCN uses two graph convolution layers to obtain image features. FuNet_C classifies the joint output features of GCN and CNN. As shown in Tables X–XIII, OA of FuNet_C is higher than that of GCN. On the SA and PU datasets, the OA of FuNet_C is 6% and 8.4% higher than that of GCN, respectively. The results show that the classification performance of both GCN and CNN is better than that of either network alone. Compared with 2-D CNN and SSRN, it is found that the accuracy of supervised methods is higher. Compared with SSCNN, it is found that the semisupervised method has lower accuracy. The results confirmed that labeled samples on model classification performance are significant.

Comparing SSDC, GCN, and FuNet_C, we find that the OA of SSDC is much higher than others. On the PU dataset, the OA of SSDC is 30.9% higher than that of GCN and 22.5% higher than that of FuNet_C. The reason is that GCN-based methods require sufficient neighborhood labeled information to classify center nodes, but GCN and FuNet_C

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TABLE XI
CLASSIFICATION ACCURACY OF ALL CATEGORIES (%), OA, AA, AND KAPPA BY DIFFERENT METHODS ON THE SALINAS DATASET

Classes	SVM	2-D CNN	SSRN	SSCNN	GCN	FuNet_C	DFSL+NN	DFSL+SVM	DCFSL	TSTnet	SSDC
1	91.40	32.04	99.73	99.68	55.16	92.10	98.29	99.75	99.21	98.76	**99.90**
2	98.17	99.00	99.65	40.62	97.63	**99.95**	98.09	99.49	99.91	98.66	99.57
3	90.90	99.63	66.10	27.99	83.72	95.97	99.18	95.54	99.33	93.10	**99.69**
4	99.06	99.83	100.00	97.97	99.06	99.85	**100.00**	99.81	99.72	99.03	99.60
5	**97.79**	95.44	64.61	94.39	76.52	94.22	96.62	96.55	96.17	91.58	96.59
6	98.56	99.78	100.00	99.48	99.93	99.98	100.00	**100.00**	99.95	98.51	99.70
7	90.17	96.42	**99.92**	69.05	99.02	99.59	99.30	98.95	99.63	98.02	99.56
8	32.53	81.72	54.72	31.72	81.66	80.17	71.39	43.21	84.60	78.57	**87.43**
9	99.14	99.87	99.40	65.65	99.12	**99.99**	97.41	98.90	99.66	99.78	99.81
10	89.17	93.39	94.59	80.07	62.29	82.83	92.58	90.94	**94.65**	86.82	93.62
11	91.60	98.10	**99.97**	55.37	78.79	94.98	97.52	98.13	99.43	97.11	99.70
12	98.49	99.95	**99.98**	47.09	65.50	80.37	99.11	99.69	99.86	99.53	99.88
13	95.71	66.18	66.26	65.92	93.76	99.22	99.11	99.59	**99.60**	99.24	99.59
14	83.13	99.21	99.65	88.81	92.97	98.29	**99.81**	99.05	99.45	98.92	99.58
15	77.94	75.43	91.89	61.23	16.42	23.13	69.46	**88.45**	80.49	77.06	87.65
16	91.55	96.80	66.63	92.40	92.66	97.58	94.80	94.39	98.24	92.35	**98.97**
OA(%)	79.64	88.65	84.35	62.44	76.74	82.78	88.45	85.19	93.42	89.57	**94.97**
AA(%)	89.08	89.55	87.69	69.84	80.89	89.89	94.54	93.90	96.87	94.00	**97.55**
Kappa($\times$100)	77.5399	87.3237	82.7253	58.7754	73.8925	80.7293	87.1740	83.6742	92.6732	88.4153	**94.3999**

TABLE XII
CLASSIFICATION ACCURACY OF ALL CATEGORIES (%), OA, AA, AND KAPPA BY DIFFERENT METHODS ON THE UNIVERSITY OF THE PAVIA DATASET

Classes	SVM	2-D CNN	SSRN	SSCNN	GCN	FuNet_C	DFSL+NN	DFSL+SVM	DCFSL	TSTnet	SSDC
1	37.36	74.93	82.24	57.29	64.59	67.86	82.79	77.95	84.64	77.41	**94.91**
2	87.46	94.51	87.85	84.92	88.18	95.22	86.03	88.46	91.19	77.26	**95.94**
3	53.09	83.10	17.67	48.47	67.98	77.62	76.45	40.22	79.47	66.00	**92.83**
4	81.75	94.24	96.99	90.67	54.89	81.09	95.22	94.42	**97.05**	94.02	95.36
5	86.12	98.02	84.98	33.33	99.22	99.95	99.82	**100.00**	99.95	99.73	99.92
6	61.23	79.33	75.81	13.85	7.64	13.55	80.82	82.47	95.84	87.68	**97.39**
7	84.22	92.66	66.64	29.79	54.98	80.54	84.78	85.75	89.59	87.87	**96.81**
8	41.96	60.95	**96.26**	31.69	17.05	25.67	73.29	82.94	83.12	66.47	93.48
9	91.02	98.56	99.68	88.30	85.29	97.20	99.00	**100.00**	99.78	99.50	98.78
OA(%)	70.54	86.36	83.03	63.11	64.83	73.26	84.68	84.28	90.28	79.75	**95.76**
AA(%)	69.36	86.26	78.68	53.14	59.98	70.97	86.47	83.61	91.18	83.99	**96.16**
Kappa($\times$100)	61.4970	82.0308	77.7526	49.8821	52.2674	63.5439	80.1290	79.5146	87.3213	74.3436	**94.4264**

only have 1.5% labeled samples. SSDC uses GAN to generate pseudo-label data of the target domain to enrich data information and get better classification results. At the same time, we find that the accuracy of SSDC is significantly improved in a slightly large class. In the IP dataset, compared with GCN and FuNet_C, SSDC improves 18.9% and 12.8% in the categories of Corn-notill and Woods, respectively. In the category of Vinyard_untrained of the SA dataset, SSDC improves 71% and 64.5%. On the PU and PC datasets, SSDC improves 84%–89.9% and 27%–30% in the category of Asphalt, respectively. This shows that we can improve the classification results of slightly large classes by enriching the training data of the target domain.

H. Compared With Few-Shot and Cross-Domain Methods

DFSL+NN and DFSL+SVM are few-shot classification methods that both adopt DFSL in the training stage and KNN and SVM, respectively, in the test stage. DCFSL is a cross-domain algorithm based on DFSL, which uses GAN to enrich the labeled data of the target domain. As shown

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TABLE XIII
CLASSIFICATION ACCURACY OF ALL CATEGORIES (%), OA, AA, AND KAPPA BY DIFFERENT METHODS ON THE PC DATASET

Classes	SVM	2-D CNN	SSRN	SSCNN	GCN	FuNet_C	DFSL+NN	DFSL+SVM	DCFSL	TSTnet	SSDC
1	98.92	**99.89**	99.40	93.41	79.00	86.63	99.65	99.85	99.52	99.66	99.75
2	86.17	94.80	64.09	13.76	86.13	89.58	92.05	92.19	93.10	94.45	**94.83**
3	83.43	94.19	92.68	85.25	**97.56**	95.31	93.32	91.53	92.52	93.23	94.93
4	71.44	**98.88**	77.09	66.62	97.11	88.75	97.65	97.69	98.30	96.10	98.78
5	75.17	87.99	**95.94**	29.06	88.96	85.05	87.16	85.16	84.21	89.01	92.78
6	72.51	97.99	**99.79**	16.38	87.63	88.44	96.50	95.52	97.77	97.77	97.59
7	92.53	87.82	**97.07**	80.11	90.99	95.43	89.10	88.00	86.72	88.86	92.47
8	70.46	98.57	96.69	72.74	**99.56**	99.34	96.54	98.13	98.34	98.52	98.85
9	89.29	99.73	**99.81**	97.04	98.56	99.75	99.07	97.79	99.49	98.77	98.17
OA(%)	86.01	97.87	96.03	74.46	87.97	91.41	96.91	97.21	97.26	97.73	**98.29**
AA(%)	82.21	95.54	91.39	61.60	91.72	92.03	94.56	93.68	94.44	95.15	**96.46**
Kappa(×100)	80.7950	96.9861	94.3978	65.1418	83.8720	88.3420	95.6440	96.0480	96.1347	96.7902	**97.5821**

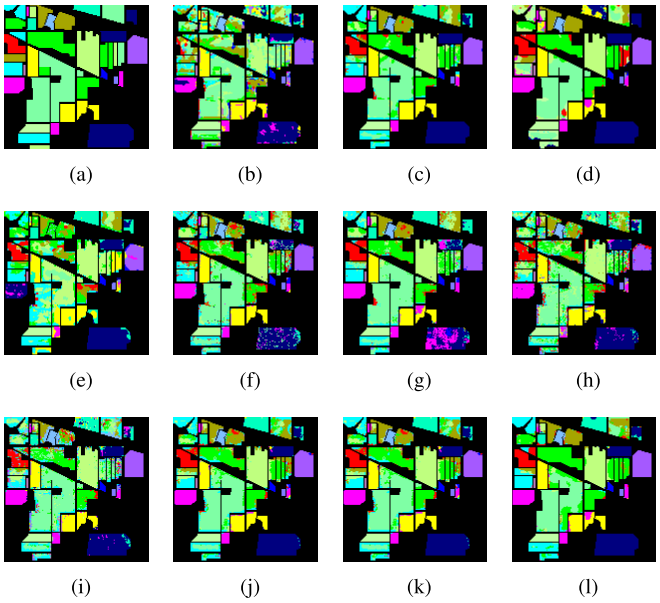

Fig. 14. Classification results obtained by all methods for the IP dataset. (a) Ground-truth. (b) SVM. (c) 2-D CNN. (d) SSRN. (e) SSCNN. (f) GCN. (g) FuNet_C. (h) DFSL+NN. (i) DFSL+SVM. (j) DCFSL. (k) TSTnet. (l) SSDC.

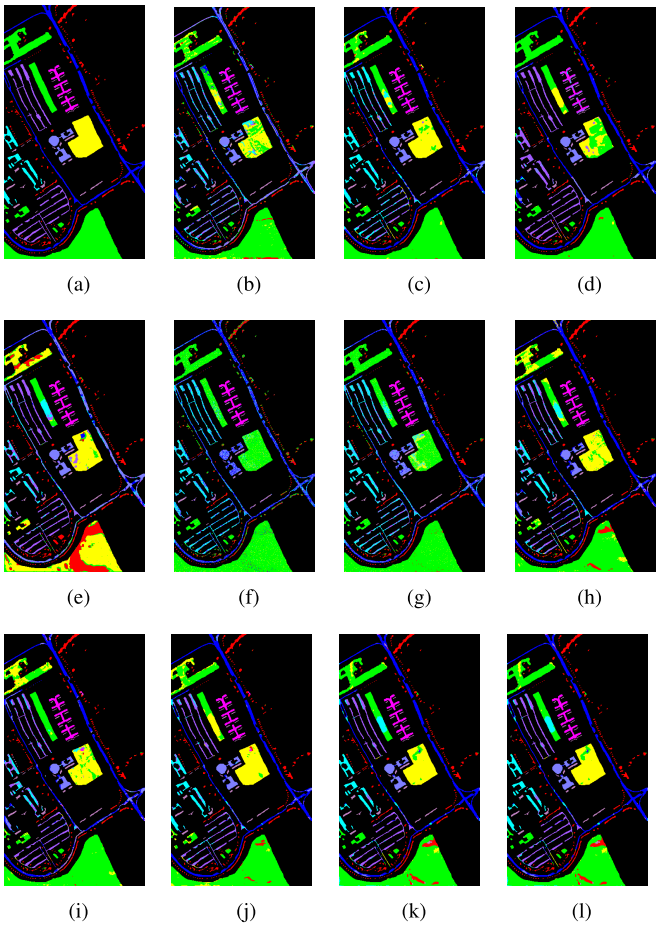

Fig. 15. Classification results obtained by all methods for the PU dataset. (a) Ground-truth. (b) SVM. (c) 2-D CNN. (d) SSRN. (e) SSCNN. (f) GCN. (g) FuNet_C. (h) DFSL+NN. (i) DFSL+SVM. (j) DCFSL. (k) TSTnet. (l) SSDC.

in Tables X–XIII, in the IP, SA, PU, and PC datasets, compared with DFSL+NN and DFSL+SVM, DCFSL increased by 10%–11%, 5%–8%, 5.6%–6%, and 0.05%–0.35%, respectively. The reason for the better classification performance of DCFSL is that it can not only learn the general features of the source domain but also extract the discriminative features of the target domain. Therefore, compared with other methods, the features extracted by DCFSL are more diverse.

Compared with the above methods, it is easy to see that SSDC greatly improves the accuracy of each category. On the IP dataset, the classification accuracy of almost every category improved by at least 2.2%. On the PU dataset, the improved range of each class is roughly 4.8%–13%. However, the performance of each category is not always the best. In the IP dataset, the accuracy of the large category of Corn-notill is 36.8%, 38.8%, and 17% higher than that of DFSL+NN,

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DFSL+SVM, and DCFSL, respectively. In the category of Corn-mintill, which contains 830 samples, SSDC is 38%, 43%, and 19.6% higher. In the category of Wheat and the small category of Stone-Steel-Towers containing 93 samples, the accuracy has decreased to about 0.64% and 1%. The results show that SSDC can obtain non-local features by using stacked cross modules, which can extract long-distance spatial information of large classes but also lose the edge information of small classes.

Therefore, SSDC uses stacked cross modules to extract the non-local spatial information of the image to compensate for the loss of edge information. In the PU dataset, SSDC is 9.9%, 7.5%, and 4.8% higher than DFSL+NN, DFSL+SVM, and DCFSL in the category of Meadows containing 18 649 samples, and 0.08%, 0.03% lower than DFSL+SVM and DCFSL in the category of sheets containing 1345 samples. Compared with the cross-domain method DCFSL, SSDC improves the OA of the IP, SA, PU, and PC datasets by 10%, 1.5%, 5.5%, and 1%, respectively. The results show that by penalizing singular values, the influence of complex channel information on classification results and the spectral data distribution shift can be reduced.

Compared with supervised, semisupervised, GCN-based, few-shot, and cross-domain methods, it can be seen that Spe can reduce spectral data distribution shift and Spa can obtain non-local spatial topology information to improve classification performance. The OA of SSDC in the IP, SA, PU, and PC datasets is 86.77%, 94,97%, 95.76%, and 98.29%. Figs. 14 and 15 visualize the classification results.

V. Conclusion

In this article, we propose SSDC, a meta-learning-based cross-domain classification network. For the image, the network introduces the attention mechanism and SVD in the spectral dimension and used the stacked cross modules and graph SAGE to extract the global spatial topology information in the spatial dimension. By parallel processing in spectral and spatial dimensions, SSDC achieves the distribution alignment of source and target domain datasets and reduces domain shift. Compared with the baseline, SSDC can obtain satisfactory classification results and has a high generalization ability.

Although DOCONV is used to replace the traditional convolution calculation in SSDC to reduce the computational complexity of the model, the time cost is still high compared with other cross-domain methods. Therefore, future research directions can be improved in modeling lightweight and more efficient GCNs.

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